Chapter 1 Introduction

Department of Statistics and Data Science

School of Economics

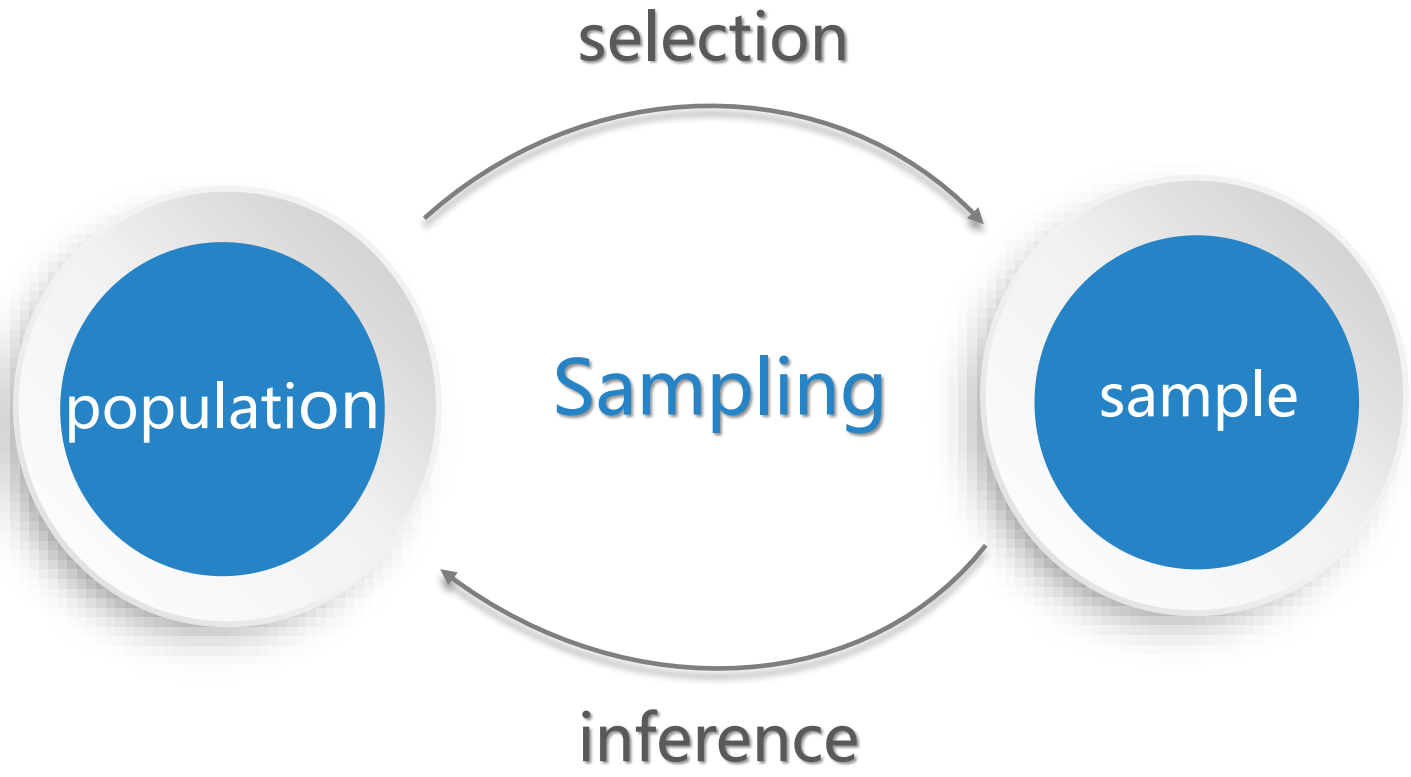
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Sampling

- The opinion of an institution that conducts thousand of transactions every day is determined by one or two encounters he has had with the institution.
- Travelers, who spend 10 days in a foreign country, then proceeded to write a book.
- The school held a symposium to collect students' opinion on teaching quality.
- …

What is sampling?

- Population
  - Finite population
  - Infinite population
- Sample: Subset of a population
- Sampling: Make statistical inference about the population without measuring the whole population
- Sampling error: The error that results from taking a sample instead of examining the whole population.


selection
population
Sampling
sample
inference

Why sampling?

- Reduced cost
- Greater speed
- Greater scope
  - Sometimes, it is the only way of getting information about the target population
- Greater accuracy

Where to use sampling?

- A prominent part in national decennial censuses
- Population survey
- Economic survey
- Market research
- Business and industry: increase efficiency
- Opinion, attitude and election polls

How to do sampling?

- Two types of sampling
  - Probability sampling
  - Non-probability sampling
- Roughly speaking, a probability rule is assigned to obtain a sample in probability sampling.
- Probability sampling is also called random sampling.

Probability sampling

- **Definition**: For each element in the population, the probability that **the element** is included in the sample is **known and greater than 0**.
- Properties
  - Able to define the set of distinct samples $S_1, S_2, \cdots, S_v$
  - Each possible sample S_i assigned a known probability of selection π_i
  - Use a random process to select one of the S_i with π_i
  - State the method for computing the estimate from the sample; a unique estimate for any specific sample



Example

Suppose that the population contains four units, numbered 1 to 4, $U = \{1, 2, 3, 4\}$.

Want to select a sample of size $n = 2$ without replacement.

$$S_1 = \{1, 2\} \qquad S_2 = \{1, 3\}$$
$$S_3 = \{1, 4\} \qquad S_4 = \{2, 3\}$$
$$S_5 = \{2, 4\} \qquad S_6 = \{3, 4\}$$

Select a sample by one of the following probability rules:

1. $\pi_i = 1/6, \quad i = 1, \ldots, 6$

2. $\pi_1 = 1/3, \pi_2 = 1/6, \pi_6 = 1/2$
$\pi_3 = \pi_4 = \pi_5 = 0$

Remarks:

1. Not all possible samples of size 2 need be included.
2. The sum of the selection probability for all possible samples equals ONE.
3. The random process used for sampling could exclude subjectivity of selecting samples.
4. We can calculate the sampling distribution of the estimates, the frequency distribution generated by the repeated application of the given sampling mechanism.
5. Sampling theory can be applied.

Nonprobability sampling

Select a sample subjectively

- Convenience sampling, accidental sampling
- Purposively sampling, judgment sampling
- Quota sampling
- Snowball sampling

Advantages: easy to understand and carry out, used to obtain preliminary information

Table 1 Quota allocations with one variable

Age	Quota
20-29	150
30-39	150
40-49	100
50+	100
Total	500
Sex	**Quota**
Male	250
Female	250
Total	500

Table 2 Quota allocations with two variables

	Male	Female	Total
20-29	70	80	150
30-39	75	75	150
40-49	55	45	100
50 +	50	50	100
Total	250	250	500

The principal steps in a sample survey

- Objectives of the survey
- Population to be sampled
- Data to be collected
- Degree of precision desired
- Methods of measurement
- The frame
- Selection of the sample
- The pretest
- Organization of the field work
- Summary and analysis of the data
- Information gained for future surveys

The role of sampling theory

Purpose: to make sampling more efficient

- Methods of sample selection
- Methods of estimation

Principle: reach specified precision at minimum cost

Use: sampling distribution,

approximately normal distribution

Concern: formulas for mean, variance, sample size

Differences

Classical sampling theory (in mathematical statistics)	Standard sampling theory (in sampling survey)
Frequency distribution of known mathematical form but unknown parameters	Very limited information about the frequency distribution, distribution free
Infinite population	Finite population

Some concepts

- Target population
  - The population about which information is wanted
  - Define population characteristics of interests (e.g. aces of corn, daily intake of calcium)
- Sampled population
  - The population to be sampled
  - Operationalize through a sampling frame
- Sampling frame
  - A list or criteria of the elements in a sampled population
  - List frame, area frame, time frame, etc.

Expect: the sampled population = the target population

Practice: the sampled population ≠ the target population

A Story

Before the 1936 US presidential election, a well-known magazine conducted a public opinion poll on the two candidates, Landon and Roosevelt. Based on the lists from the phone book and vehicle registration, a total of 10 million questionnaires were sent out, and 2.4 million were returned.

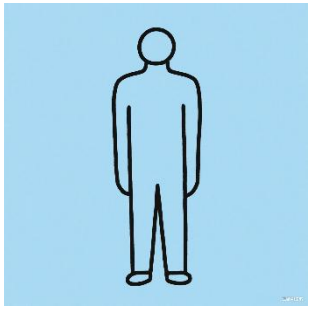
The survey results predicted that the Republican candidate Landon would defeat the Democratic candidate Roosevelt with 57% to 43%, but the actual result was 38% to 62%.

Some concepts

- Sampling frame error
  - Match between the population and sampling frame
- Sampling unit
  - Unit to be sampled
  - For example, individual, family, organization
- Unit / reporting unit / survey unit
  - The element, the observational unit

The sampling unit can be a group of survey units.







Example

For the following survey, describe the target population, sampling frame, sampling unit, reporting unit. Discuss any possible sources of selection bias or inaccuracy of responses.

A sample of 8 architects was chosen in a city with 14 architects and architectural firms. To select a survey sample, each architect was contacted by telephone in order of appearance in the telephone directory. The first 8 agreeing to be interviewed formed the sample.

Ans:

Target population: all 14 architects in the city

Sampling frame: the telephone directory for the 14 architects

Sampling unit: each phone number in the telephone directory

Reporting unit: each architect

Definitions and notations

- $U = \{1, 2, \ldots, N\}$: index set of finite population
- $S_1, S_2, \cdots, S_v$: all possible samples
- $\pi_i = P(S_i)$: the probability to select S_i, $\sum_{i=1}^{v} \pi_i = 1$
- $\hat{\theta} = \hat{\theta}(S)$: the estimator of some population characteristic θ , a statistic
- $\hat{\theta}_i = \hat{\theta}(S_i)$: the estimate given by the ith sample

With probability distribution of samples, or sampling distribution $\pi_i, \quad i = 1, \ldots, v$, probability distribution of a statistics is induced.

- Expectation: $E(\hat{\theta}) = \sum_{i=1}^{v} \pi_i \hat{\theta}_i$
- Variance: $\mathrm{Var}(\hat{\theta}) = \sum_{i=1}^{v} \pi_i (\hat{\theta}_i - E(\hat{\theta}))^2$
- Mean square error:

$$\begin{aligned} \mathrm{MSE}(\hat{\theta}) &= \sum_{i=1}^{v} \pi_i (\hat{\theta}_i - \theta)^2 \\ &= \mathrm{Var}(\hat{\theta}) + [E(\hat{\theta}) - \theta]^2 \end{aligned}$$

- Bias: $B = E(\hat{\theta}) - \theta$
- The estimator $\hat{\theta}$ is called unbiased if $B = 0$.
- What is sampling error ?

(sampling) error of $\hat{\theta}$

$$\begin{aligned} &= \hat{\theta} - \theta \\ &= \{\hat{\theta} - E(\hat{\theta})\} + \{E(\hat{\theta}) - \theta\} \\ &= \text{variation} + \text{bias} \end{aligned}$$

! Probability sampling: there exists variation but bias is usually 0 or close to 0.



Example (continued)

A population $U = \{1, 2, 3, 4\}$ → a sample of $n = 2$

Sample	Unit ID	sample mean $\bar{y}$	sampling error	Selection probability
S_1	1, 2	3/2	-1	1/6
S_2	1, 3	2	-1/2	1/6
S_3	1, 4	5/2	0	1/6
S_4	2, 3	5/2	0	1/6
S_5	2, 4	3	1/2	1/6
S_6	3, 4	7/2	1	1/6

Here the sample mean has a direct probability distribution.

- Probability distribution of the sample mean $\bar{y}$, or the sampling distribution:

$$P_{\bar{y}}(y) = \begin{cases} 1/6, & \text{if } y \in \{3/2,\ 2, 5/2, 5/2, 3, 7/2\} \\ 0, & \text{else} \end{cases}$$

- Unbiased

- Variance

In practice, the exact sampling error is unknown…

Sampling error and large number theory

- Sampling error is measured by standard error of $\hat{\theta}$

$$\sigma_{\hat{\theta}} = \sqrt{\mathrm{Var}(\hat{\theta})}$$

- **Large number theory**

  Law of Large Numbers: $\hat{\theta}$ converges to $E(\hat{\theta})$ for sufficiently large size.

  Central Limit Theorem: $\hat{\theta}$ follows a normal distribution for sufficiently large size.



Example

A population contains eight units, numbered 1 to 8. Denote the value of the ith unit by y_i. The population is $U = \{1, 2, 3, 4, 5, 6, 7, 8\}$

The values of all units are

i	1	2	3	4	5	6	7	8
y_i	1	2	4	4	7	7	7	8

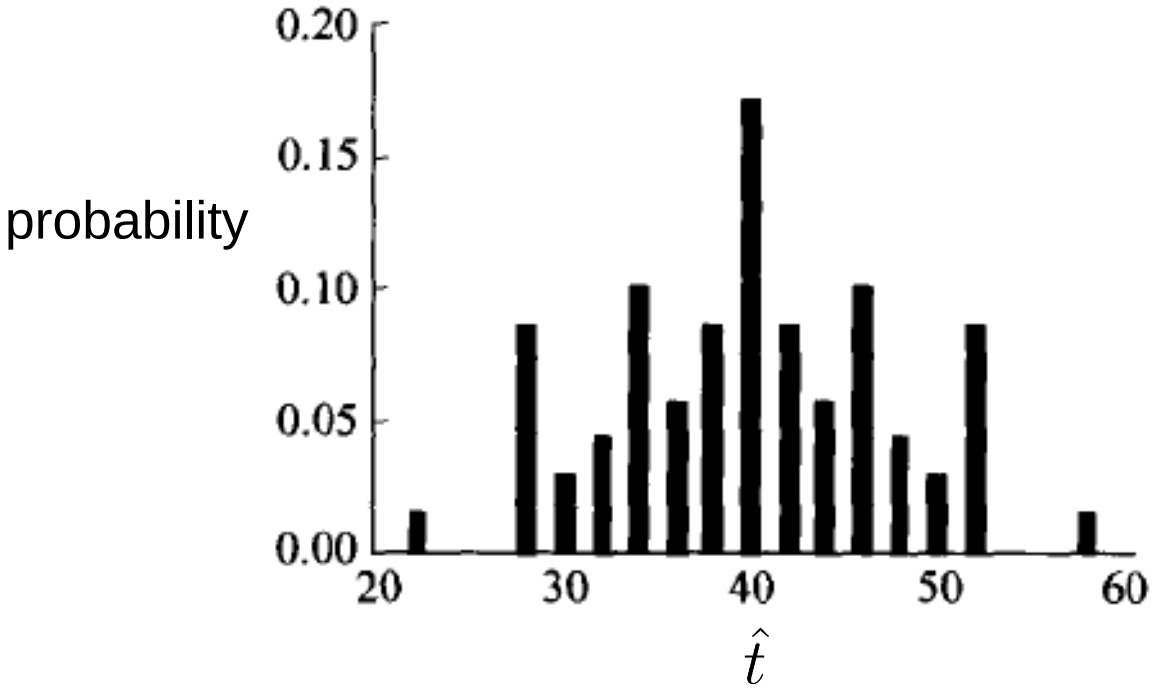
Simple random sampling $n = 4$; denote estimator of total by $\hat{t}$.

E.g. there are 4 samples with a same sample total 22. So the probability that $\hat{t}$ takes value 44 is 4/70

$$\{3, 4, 5, 6\}, \{3, 4, 5, 7\}, \{3, 4, 6, 7\}, \{1, 5, 6, 7\}$$

- Sampling distribution

k	22	28	30	32	34	36	38	40	42	44	46	48	50	52	58
$P\{\hat{t} = k\}$	$\frac{1}{70}$	$\frac{6}{70}$	$\frac{2}{70}$	$\frac{3}{70}$	$\frac{7}{70}$	$\frac{4}{70}$	$\frac{6}{70}$	$\frac{12}{70}$	$\frac{6}{70}$	$\frac{4}{70}$	$\frac{7}{70}$	$\frac{3}{70}$	$\frac{2}{70}$	$\frac{6}{70}$	$\frac{1}{70}$



Use of normal distribution

To compute confidence interval

- To find two statistics $\hat{\theta}_U, \hat{\theta}_L$ such that $P(\hat{\theta}_L \le \theta \le \hat{\theta}_U) = 1-\alpha$.
- $[\hat{\theta}_L, \hat{\theta}_U]$ is a confidence interval for θ with a confidence level $1-\alpha$. And the limits $\hat{\theta}_L$ and $\hat{\theta}_U$ are called lower and upper confidence limits.

Use pivotal statistics to get the CI

According to CLT: for sufficiently large size, $\hat{\theta} \sim N(\theta, \sigma^2_{\hat{\theta}})$.

$$\Rightarrow \frac{\hat{\theta}-\theta}{\sigma_{\hat{\theta}}} \sim N(0,1) \qquad \Rightarrow P\left(-z_{\alpha/2} \le \frac{\hat{\theta}-\theta}{\sigma_{\hat{\theta}}} \le z_{\alpha/2}\right) \doteq 1-\alpha$$

$$\Rightarrow P\left(\hat{\theta} - z_{\alpha/2}\sigma_{\hat{\theta}} \le \theta \le \hat{\theta} + z_{\alpha/2}\sigma_{\hat{\theta}}\right) \doteq 1-\alpha$$



Example

If a probability sample of the records of batteries in routine use in a large factory shows an average life days $\hat{\mu} = 394$, with a standard error $\sigma_{\hat{\mu}} = 4.6$ days.

The chances are 99% that the average life in the population of batteries lies between

$$\hat{\mu}_L = 394 - 2.58 \times 4.6 = 382 \text{ (days)}$$

and

$$\hat{\mu}_U = 394 + 2.58 \times 4.6 = 406 \text{ (days)}$$

The two limits are lower and upper confidence limits.

$z_{0.16}$	$z_{0.05}$	$z_{0.025}$	$z_{0.005}$
1	1.64	1.96	2.58

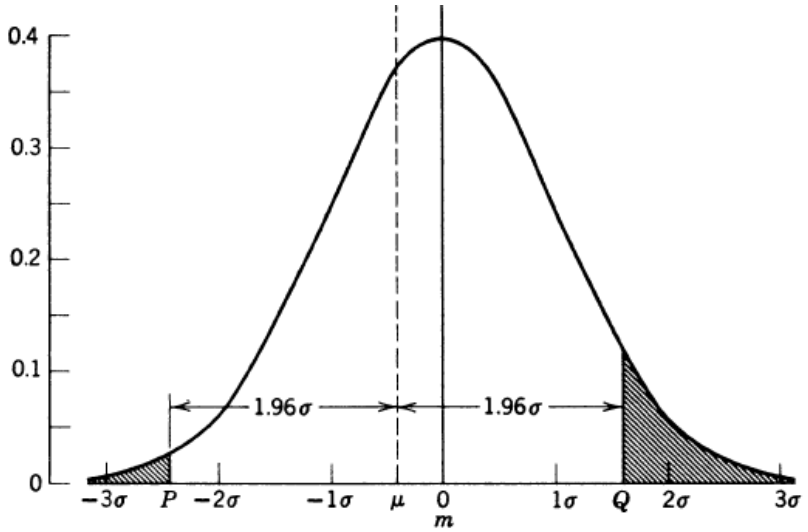
- With a single estimate from a single survey, the statement “ μ lies between 382 and 406 days” is not certain to be correct.

- What does the “99% confidence” mean?
- **Answer:** if the same sampling plan were used many times in a population, a confidence statement being made from each sample, about 99% of these statements would be correct and 1% wrong.

Bias and its effects

- **If there is a bias** $E(\hat{\mu}) = m \neq \mu$

$P(\hat{\mu} < \mu - 1.96\sigma)$



$P(\hat{\mu} > \mu + 1.96\sigma)$

Fig. 1.1 Effect of bias on errors of estimation.

Interested in the probability $P(|\hat{\mu} - \mu| \leq 1.96\sigma)$

$\Leftrightarrow$ $P(\hat{\mu} < \mu - 1.96\sigma) + P(\hat{\mu} > \mu + 1.96\sigma)$

$$P(\hat{\mu} < P) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{-1.96-(B/\sigma)} e^{-t^2/2} dt$$

$$P(\hat{\mu} > Q) = \frac{1}{\sqrt{2\pi}} \int_{1.96-(B/\sigma)}^{\infty} e^{-t^2/2} dt$$

	Probability of Error		
B/σ	$< -1.96\sigma$	$>1.96\sigma$	Total
0.02	0.0238	0.0262	0.0500
0.04	0.0228	0.0274	0.0502
0.06	0.0217	0.0287	0.0504
0.08	0.0207	0.0301	0.0508
0.10	0.0197	0.0314	0.0511
0.20	0.0154	0.0392	0.0546
0.40	0.0091	0.0594	0.0685
0.60	0.0052	0.0869	0.0921
0.80	0.0029	0.1230	0.1259
1.00	0.0015	0.1685	0.1700
1.50	0.0003	0.3228	0.3231

Outline of a sample survey

1. Define a target population
2. Determine population characteristics of interest
3. Find sampling frame: device that associates elements by sampling units (e.g. phone book)
4. Obtain a sample by a probability sampling design
5. Measure the study variables
6. Use measured values to compute point estimates, standard errors, and confidence intervals

Sources of errors

- Sampling error:
  - If $n = N$, no sampling error
  - In fact, if $n\uparrow$ then sampling error $\downarrow$
- Nonsampling error: Every type of errors else
  - Such as measurement error, processing error, etc.
  - Even if $n = N$, we have nonsampling error.
  - In practice, we can decrease nonsampling error by decreasing n .
- Because of nonsampling error, a sample is often more accurate than a census.

Total error

Summary

The course is about the sampling error of $\hat{\theta}$.

1. What is it?
2. How to reduce it?
3. How to measure it?